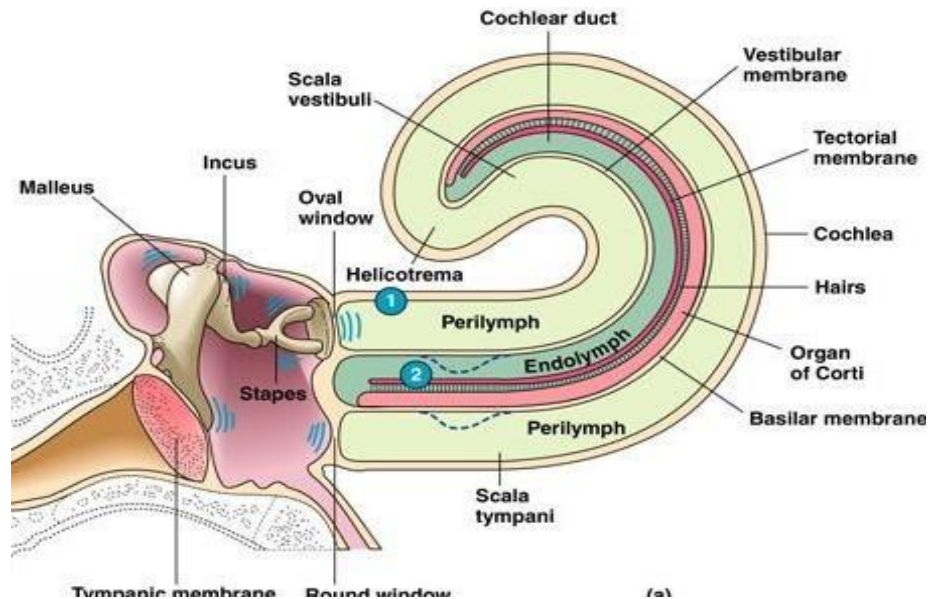


receptors called **maculae** of the utricle and saccule which detect **linear acceleration** of the head in the vertical and horizontal planes.

Three semi-circular ducts:

- They are located within the semi-circular canals.
- They contain sensory receptors (**3 cristae ampullaris**) in the ampullae of the semi-circular canals for kinetic balance of head movement (sensitive to angular acceleration of the head).



Blood supply of the labyrinth:

- **Labyrinthine artery** from the basilar artery.
- **Stylomastoid artery** from posterior auricular artery.

Lecture (11): Spinal cord I

Gross anatomy of the Spinal Cord:

• Position:

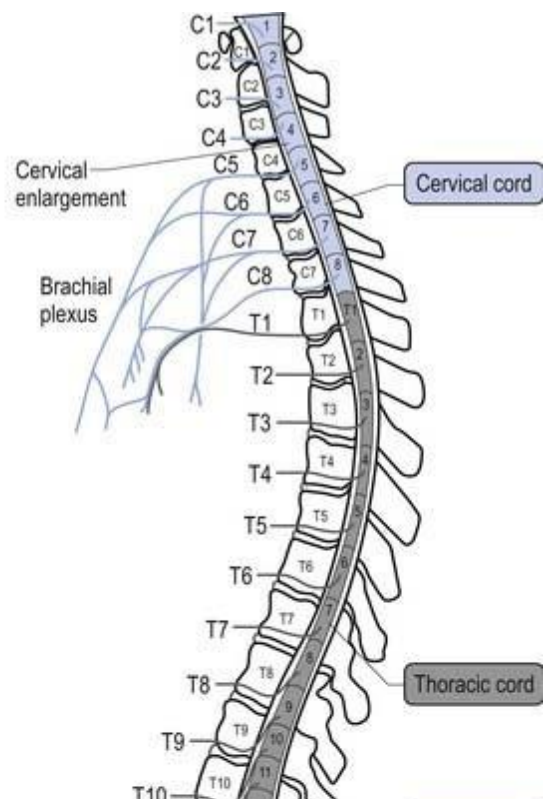
The spinal cord connects the brain to the peripheral nervous system (PNS). It is the caudal part of the central nervous system that occupies the upper two-thirds of the vertebral canal.

• Shape:

The spinal cord is roughly cylindrical in shape, 45cm long.

• Beginning:

It begins superiorly at the foramen magnum in the skull base, where it is continuous with the medulla



oblongata.

- **Termination:**

- The adult spinal cord ends inferiorly at the level of the disc between **L1 & L2 vertebrae**.
- The lower tapering end of spinal cord is called the conus medullaris.
- At the **3rd intratrauterine month**, it occupies the whole length of the vertebral canal.
- At birth, its lower end lies opposite **L3 vertebra**.

- **It has two fusiform enlargements:**

1- **Cervical enlargement** where it gives origin to the brachial plexus.

2- **Lumbar enlargement** where it gives origin to the lumbosacral plexuses.

- **Ventrally**, the spinal cord possesses a deep longitudinal fissure called anterior median fissure.
- **Dorsally**, the spinal cord shows a shallow furrow called the posterior median sulcus.
- **Two anterolateral sulci**; along which the ventral roots leave the spinal cord.
- **Two posterolateral sulci**; along which the dorsal roots enter the spinal cord.

- **Protection of spinal cord:**

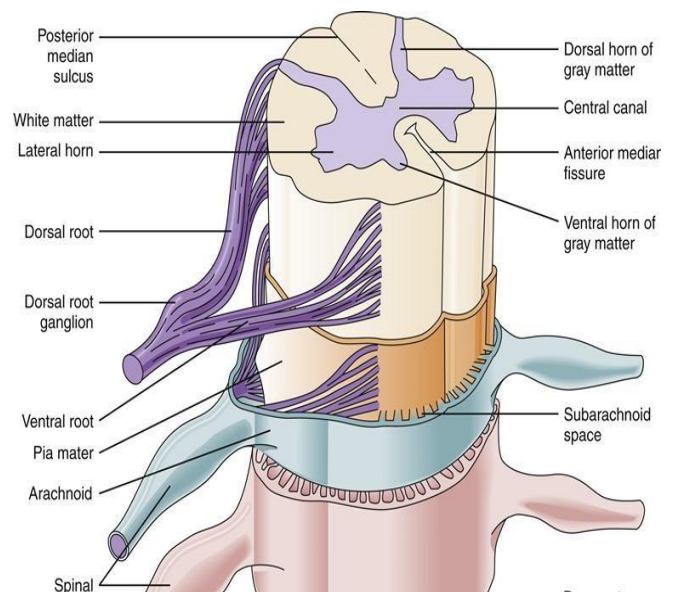
1- **The vertebral canal and the dura mater** provide structural protection.

2- **Filum terminale**: downward prolongation of pia mater extends from the conus medullaris to back of coccyx.

3- **Denticulate ligaments**: lateral extensions of pia mater to attach dura mater.

4- **The cerebrospinal fluid.**

- **Spinal nerve roots:**



- **31 pairs of spinal nerves** along the entire length of the spinal cord are attached by two roots; **the ventral motor root and the dorsal sensory root.**
- Each dorsal nerve root possesses a dorsal root ganglion, the cells of which give rise to peripheral and central nerve fibers.
- The ventral and dorsal roots of the spinal nerves unite within the intervertebral foramina to form **spinal nerve.**
- Lumbosacral nerve roots (**L2-S5 and coccygeal**) are suspended from the lower end of the spinal cord, forming a bundle of nerves known as the **cauda equina (horse tail-like).**

Blood supply of spinal cord:

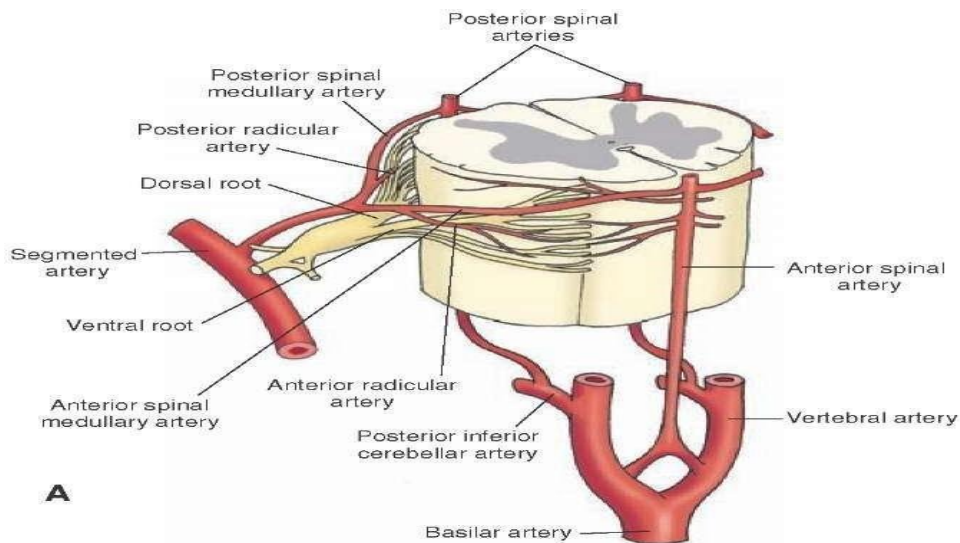
Arterial supply:

- 1- Anterior spinal artery** arises by two branches from Rt & Lt vertebral arteries. It descends in anterior median fissure. It supplies anterior two-thirds of the spinal cord segment.
 - 2- Rt & Lt posterior spinal arteries:** Each arises from vertebral artery. It descends in posterolateral sulcus. It supplies posterior one-third of the spinal cord segment.
 - 3- Radicular spinal arteries:** they arise regionally from vertebral, intercostal, lumbar and lateral sacral arteries. They enter the vertebral canal through intervertebral foramina to supply the lower two-thirds of the spinal cord.
 - 4- Great anterior radicular artery of Adamkiewicz (arteria radicularis magna)** arises from the left lower intercostal or upper lumbar arteries, and is the major blood supply of lumbar and sacral regions of the spinal cord.
- **Occlusion of the anterior spinal artery** may lead to the (**Anterior Cord syndrome**):

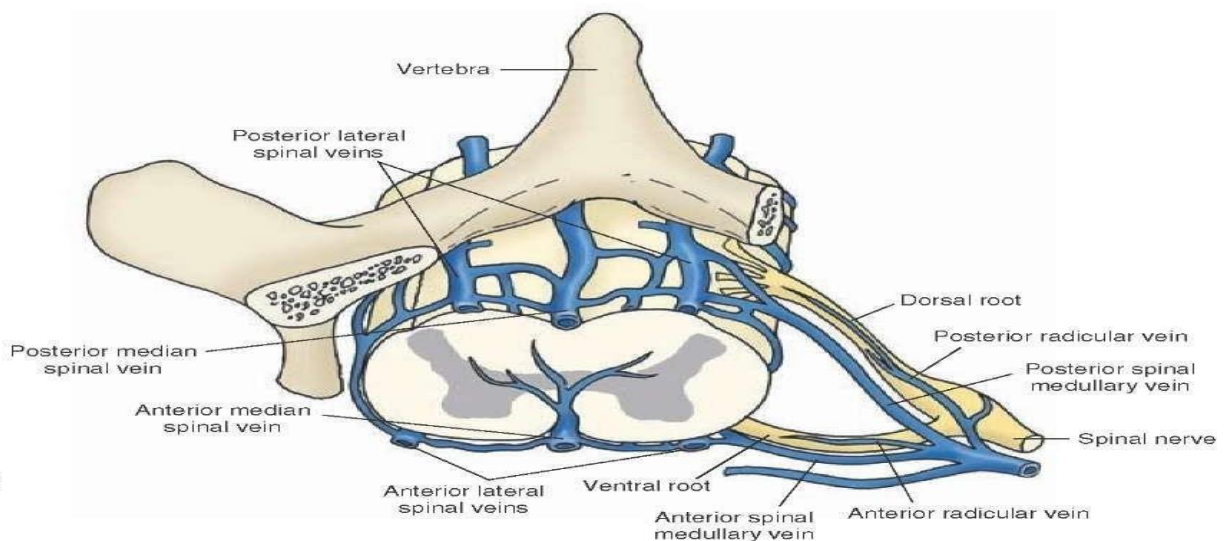
- Loss of ipsilateral motor function, due to damage to ventral gray matter and the ventral corticospinal tract.
- Loss of contralateral pain and temperature sensation, due to damage of lateral spinothalamic pathway. Paraplegia and incontinence.

Venous drainage:

- They are **3 anterior spinal veins** run along the anterior median fissure and ventral roots and, **3 posterior spinal veins** run along posterior median sulcus and dorsal roots.
- These veins form an anastomosing network along the surface of the spinal cord. The venous channels communicate superiorly with the **cranial dural venous sinuses** and laterally with the **internal vertebral venous plexuses**, which in turn empty into systemic segmental veins.



A



B

Internal structure of the Spinal Cord:

- The spinal cord is divided into **31 segments** as follows:
8 cervical, 12 thoracic, 5 lumbar, 5 sacral and one coccygeal segments.
- **Each spinal cord segment** is divided into Rt & Lt similar halves by **anterior median fissure & posterior median sulcus**.
- The two halves are connected by: **White commissure** in front of the central canal.
- Grey commissure is the central limb of the H-shaped grey matter in cross-section and contains the central canal.

The spinal cord segment: is composed of:

- A central mass of **grey matter** (cell bodies), which is reminiscent of the shape of butterfly wings (H-shaped) around the central canal.
- Outer cylindrical mass of **white matter** is formed of myelinated nerve fibers.

Grey matter of spinal cord segment: consists of:

- **Two anterior (ventral) horns:** contain motor neurons called anterior horn cells (AHC) whose axons exit the cord via the ventral root, on their way to the muscles.
- **Two posterior (dorsal) horns:** contain sensory neurons called posterior horn cells (PHC).
- **Two lateral horns:** in certain segments, containing autonomic neurons.

Nuclei of the ventral horn:

- There are two structural types of motor neurons:
 - 1- **Alpha α -motor neurons:** supply extrafusal muscle fibers.
 - 2- **Gamma γ -motor neurons:** supply intrafusal muscle fibers (muscle spindles).
- The motor neurons are arranged into three basic groups:
 - 1- **Medial group:** innervates axial muscles (trunk musculature), presents in all spinal cord segments.
 - 2- **Lateral group:** Innervates limb muscles, presents mainly in cervical, lumbosacral segments.
 - 3- **Central group:**
 - Innervates the diaphragm, presents in 3rd, 4th, 5th cervical segments, referred as phrenic nucleus.
 - Innervates sternomastoid & trapezius, presents in upper 5 or 6

cervical segments and is referred as accessory nucleus.

Nuclei of the dorsal horn:

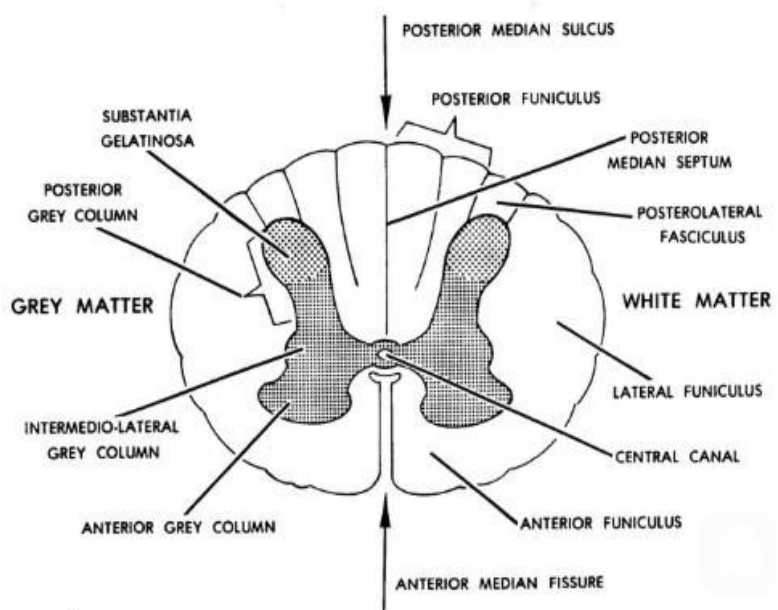
- 1- **Substantia Gelatinosa of Rolandi (SGR):** It lies at top of dorsal horn throughout the length of the spinal cord. It is associated with pathway of pain & temperature sensations.
- 2- **Nucleus proprius:** It lies at the middle of dorsal horn, ventral to SGR throughout the length of the spinal cord. It is associated with crude touch pathway.
- 3- **Clarke's nucleus (Nucleus dorsalis):** It lies at the base of dorsal horn at Level: C8 – L3 segments of the spinal cord, it receives unconscious proprioception from the lower limb & projects to cerebellum.
- 4- **Visceral afferent nucleus:** It is situated lateral to the nucleus dorsalis, it extends from T8 – L3 segments of the spinal cord. It is associated with receiving visceral sensation.

Nuclei of the lateral horn:

- 1- **Preganglionic sympathetic cell bodies** are present at level of T1 – L2 segments of spinal cord.
- 2- **Preganglionic parasympathetic cell bodies** are present in S2, 3, 4 segments (no lateral horn).

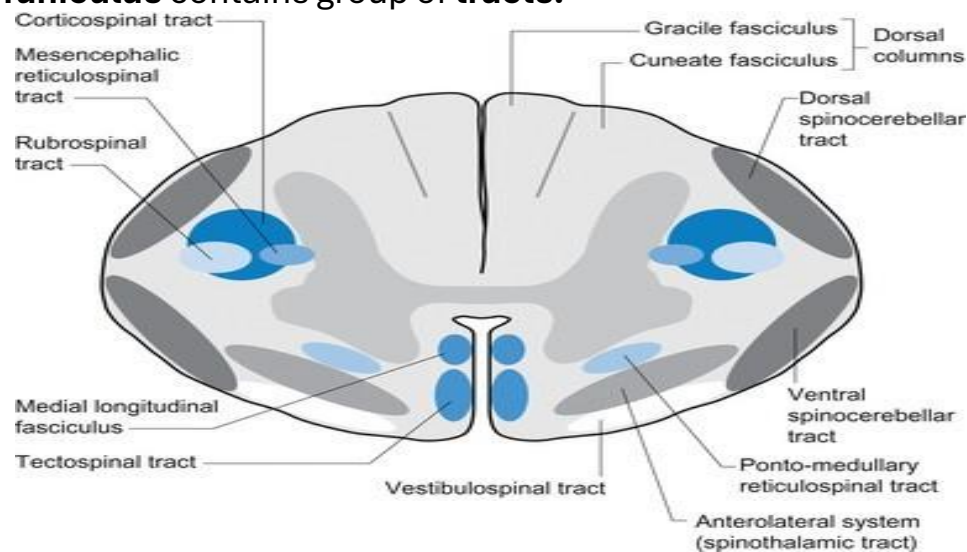
White matter of spinal cord segment:

- The **white matter** is organized into anterior, lateral, and posterior white columns or funiculi.
- The **anterior white column** on each side lies between the midline and the exit of the anterior roots.
- The **lateral white column** lies between the exit of anterior roots and the entry of the posterior roots.
- the **posterior white**



column lies between the entry of the posterior roots and the midline.

- **Each funiculus** contains group of **tracts**.



Lecture (12): Spinal cord II

- The **spinal tracts** are divided into:
 - Ascending sensory tracts.
 - Descending motor tracts.
 - Intersegmental short tracts

Ascending sensory tracts:

- The posterior white column is wholly occupied by the **ascending sensory tracts**.
- The main ascending tracts of lateral white column are the **lateral spinothalamic tract, anterior & posterior spino-tectal tracts**.
- The main ascending tracts of anterior white column is the **ventral spinothalamic tract**.
- The sensation from the sensory receptors to the cerebral cortex or subcortical is conducted by series of neurons called pathway.

